Executive Analytics & Quality Briefing

ISB Capstone Project | Generated: 2026-09-08 16:03:06

1. Executive Summary

This strategic intelligence report provides an in-depth audit of South Region transaction records across 82,034 historical sales rows and 23 operational dimensions. The dataset demonstrates enterprise-grade readiness with a 94.99% data completeness rating.

Key Performance Indices:

- Total Transaction Rows: 82,034
- Data Completeness Score: 94.99%
- Total Missing Value Cells: 94,435 (5.01%)
- Duplicate Records Identified: 4,463 (5.44%)
- Columns Requiring Standardization: 23 of 23

2. Operational Data Diagnostics

Column-Level Missing Profile:

- Material: 1 missing (0.00%)
- Billing Doc.: 23,604 missing (28.77%)
- Item: 23,604 missing (28.77%)
- SALES OFFICE CODE: 1 missing (0.00%)
- EXT MAT GROUP: 1 missing (0.00%)
- MatGroup3: 1 missing (0.00%)
- Star Rating: 1 missing (0.00%)
- Segment: 1 missing (0.00%)
- Star Rating.1: 1 missing (0.00%)
- Stor. Location: 23,604 missing (28.77%)
- SALES OFFICE CODE.1: 1 missing (0.00%)
- SBU: 1 missing (0.00%)
- Year: 1 missing (0.00%)
- Month: 1 missing (0.00%)
- Week: 1 missing (0.00%)
- Billing Date: 1 missing (0.00%)
- Customer: 1 missing (0.00%)
- Material.1: 1 missing (0.00%)
- Billing Quantity ODU: 1 missing (0.00%)
- Billing Quantity IDU: 1 missing (0.00%)
- Tonnage: 1 missing (0.00%)
- Dealer Name: 23,604 missing (28.77%)
- Tonnage.1: 1 missing (0.00%)

3. Strategic Recommendations

1. Demand Forecasting Pipeline:

- Deploy SKU-level models across Bangalore, Chennai, Cochin, Secunderabad, and Vijayawada.
- Employ SARIMA for high-seasonality items and Gradient Boosting for multi-feature velocity.

2. Data Governance:
- Dedup and prune redundant sales records.
- Establish automated weekly data quality validation checks.